IPML-WS2026-27-Exercises

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1 Introduction & From Precise to Imprecise Probabilities

1.1 False Confidence in Decision-Making

Consider an agent who must decide whether to take an umbrella before going out. The agent's payoffs are given in the following table:

		Weather	
		Rain	No rain
Action	Umbrella	0	-1
	No umbrella	-2	2

Two forecasters are asked to predict tomorrow's weather in a region about which they have no prior information. The first one nevertheless assigns a precise probability of 0.5 for rain, while the second reports a probability interval of $[0.0, 1.0]$ to reflect their uncertainty.

Suppose the agent trusts the first forecaster,

(a) What is their expected utility for each action?

(b) What is the optimal action, i.e., one that maximises the expected utility? If it rains tomorrow, what payoff does the agent receive?

If the agent trusts the second forecaster,

(c) What is the lower expected utility (i.e., the minimum expected utility over all probability distributions compatible with the forecast) for each action? What is the upper expected utility for each action?

(d) Given the lower and upper expected utilities of each action as computed previously, how would you proceed if you were the agent?

1.2 Ambiguity Aversion

Consider two urns, A and B, each containing red and blue balls. Urn A contains 50 red and 50 blue balls, while the proportions of red and blue balls in urn B are unknown. A player must choose an urn from which a ball will be drawn at random. The player exhibits the following preferences:

- If drawing a red ball yields €1 and drawing a blue ball yields €0, the player prefers urn A over urn B.
- If drawing a blue ball yields €1 and drawing a red ball yields €0, the player also prefers urn A over urn B.

Consider the subjective interpretation of precise probabilities,

(a) Consider the two indicator gambles, $\mathbf{1}_r^A$ and $\mathbf{1}_r^B$, which equal 1 if a red ball is drawn from the respective urn and 0 otherwise. Let p_r^A and p_r^B denote the player's fair prices for these gambles, which fair price is higher?

(b) What does this imply about the player's *subjective* probability of drawing a red ball from each urn?

(c) What about the player's *subjective* probability of drawing a blue ball from each urn?

(d) If $p_r^A + p_b^A = 1$, what is $p_r^B + p_b^B$? Is this consistent with the additivity of precise probabilities?

Suppose instead that the player represents their uncertainty about drawing a red ball from urn B using the probability interval $[0.0, 1.0]$.

(e) What are the lower and upper probabilities $(\underline{p}_r^B, \overline{p}_r^B)$ of the event "*drawing a red ball*" from urn B?

(f) What are the lower and upper previsions $\underline{P}(1_r^B), \overline{P}(1_r^B)$ of the gamble $\mathbf{1}_r^B$?

(g) Use the conjugacy rule $\overline{p}(e) = 1 - \underline{p}(e^c)$ (for any event e), compute the lower and upper previsions $\underline{P}(1_b^B), \overline{P}(1_b^B)$ of the gamble $\mathbf{1}_b^B$?

(h) If the player uses the Γ-maximin decision rule, i.e., choosing the urn that has the highest lower prevision, which urn will they choose if they are rewarded for drawing a red ball? What about a blue ball?

2 Probability Refresher

2.1 Common Pitfalls

The statements below are common in writing but ambiguous. For each one, name the confusion.

(a) "A random variable is a variable whose outcome is random."

(b) "Let X be a Gaussian distribution with mean 0."

(c) "Every subset of Ω has a probability."

(d) "X is continuous, so $\mathbb{P}(X = 0.3) = f_X(0.3)$."

(e) "This event has probability zero, so it will never happen."

2.2 Kolmogorov's Model of Probability

Toss a fair coin twice. Take $\Omega = \{HH, HT, TH, TT\}$, let $\mathcal{F}$ be the set of all subsets of Ω, and let $\mathbb{P}(\{\omega\}) = \frac{1}{4}$ for every $\omega \in \Omega$. Let $X(\omega)$ be the number of heads in ω.

(a) Fill in the values of X. Then fill in the preimage $X^{-1}(B)$ and the probability $P_X(B)$ for each set B in the second table.

ω	HH	HT	TH	TT
$X(\omega)$				

B	$X^{-1}(B)$	$P_X(B)$
$\{0\}$		
$\{1\}$		
$\{1,2\}$		
$(-\infty, 1.5]$		
$\{5\}$		

(b) Write down the PMF p_X. Then give $F_X(t)$ for every $t \in \mathbb{R}$ and sketch it.

(c) For each object, state what it takes as input and what it returns. The first row is done for you.

Object	Input	Output
X	an outcome $\omega \in \Omega$	a real number $X(\omega)$
X^{-1}		
$\mathbb{P}$		
P_X		
p_X		
F_X		

(d) First, consider a fair die whose outcome we only partly observe. Let $\Omega = \{1, \ldots, 6\}$ and suppose we only learn whether the roll is even or odd, so that

$$\mathcal{F} = \big\{\emptyset, \{2,4,6\}, \{1,3,5\}, \Omega\big\}, \qquad \mathbb{P}(\{2,4,6\}) = \mathbb{P}(\{1,3,5\}) = \tfrac{1}{2}.$$

Is $X_1(\omega) = \omega \bmod 2$ a random variable on $(\Omega, \mathcal{F})$? Is $X_2(\omega) = \omega$? Can you compute $\mathbb{P}(\{6\})$ in this model?

3 Betting Interpretation of Probability

Show that

1. A pricing scheme that is not finitely additive is not coherent.
2. Prove that a finitely additive pricing scheme is coherent.